

Electricity flexibility in Kerala

Import dependence, hydropower and demand timing

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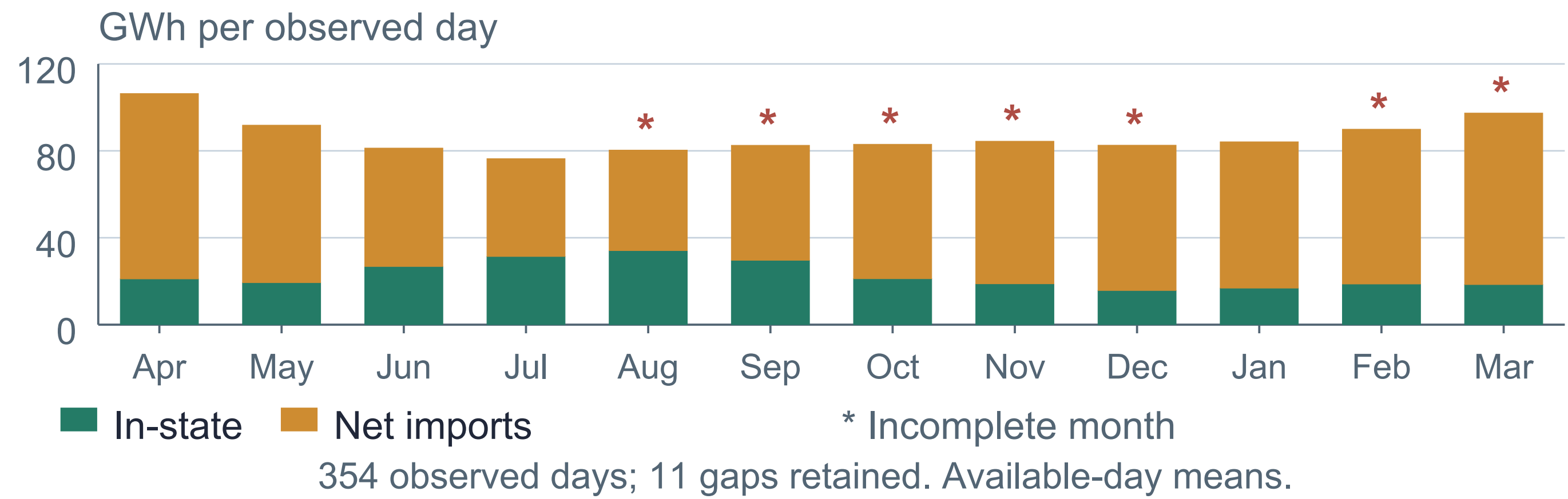
Project description

Context and objective

Kerala's FY2024-25 electricity record is **74% net imports**. Hydropower supplies most in-state generation.

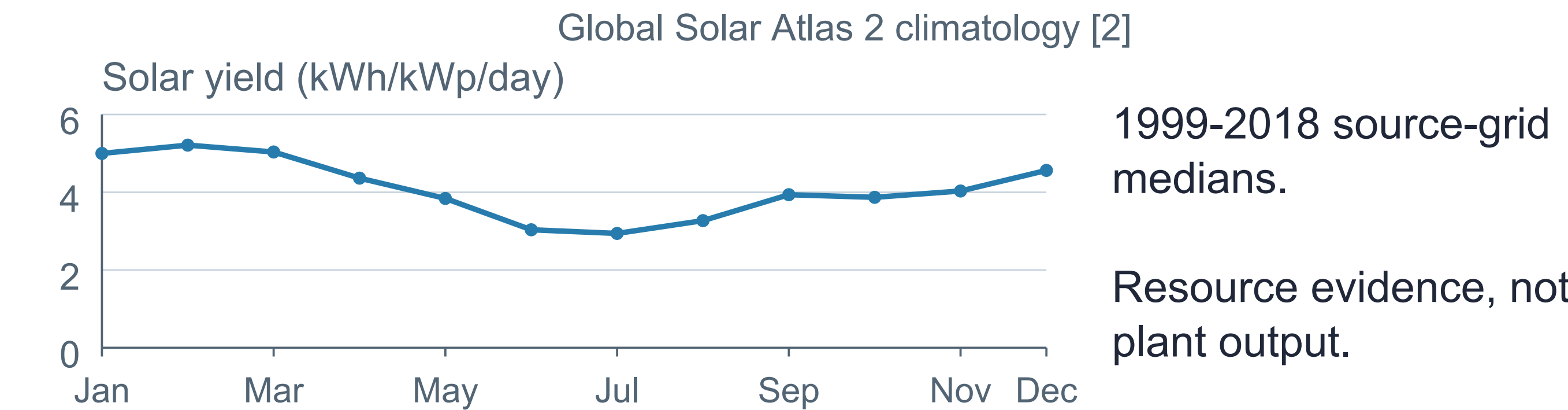
Objective: test how import constraints, hydro operation and demand timing affect electricity adequacy.

A. Observed electricity balance [1]

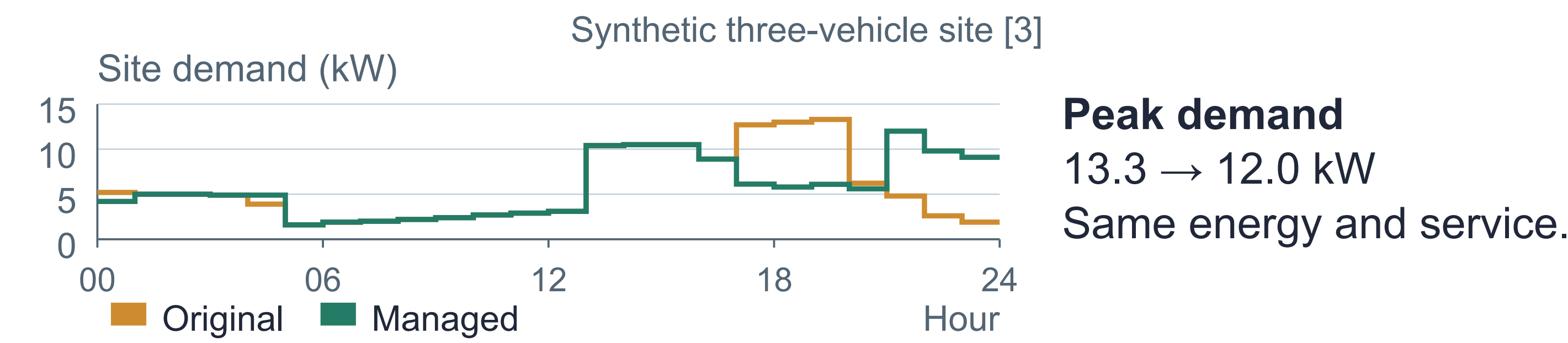


Resources and technologies

B. Seasonal solar resource

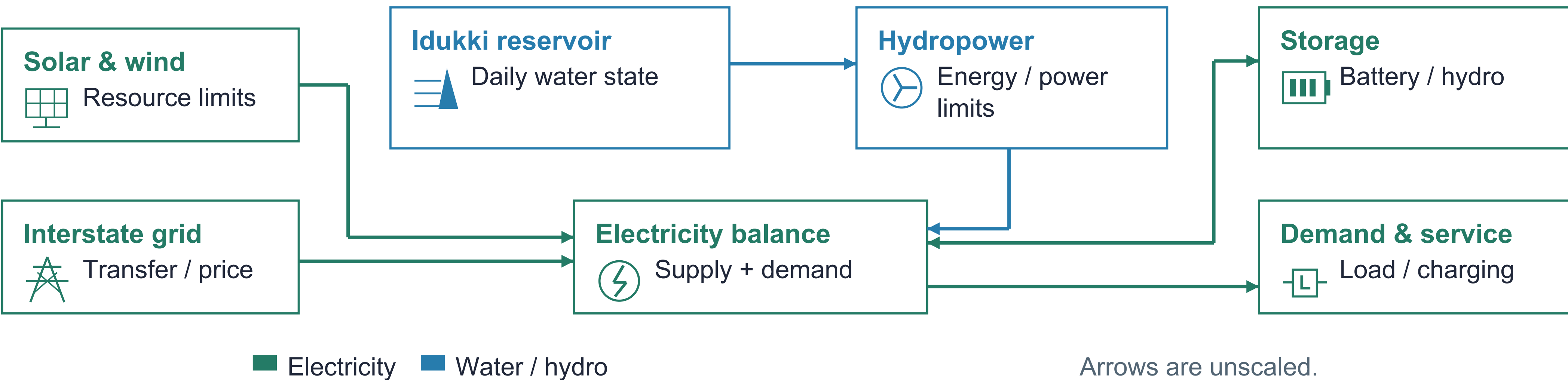


C. Managed vehicle charging



System framework and methods

Linked research modules



Scope: separate model experiments, not a validated statewide capacity plan. Land, ecology and industrial integration remain future work.

Electricity balance

$G + M + P_{dis} + U = D + P_{ch}$
G generation; M imports; D demand; U unmet load;
P storage charge/discharge.

Daily reservoir balance

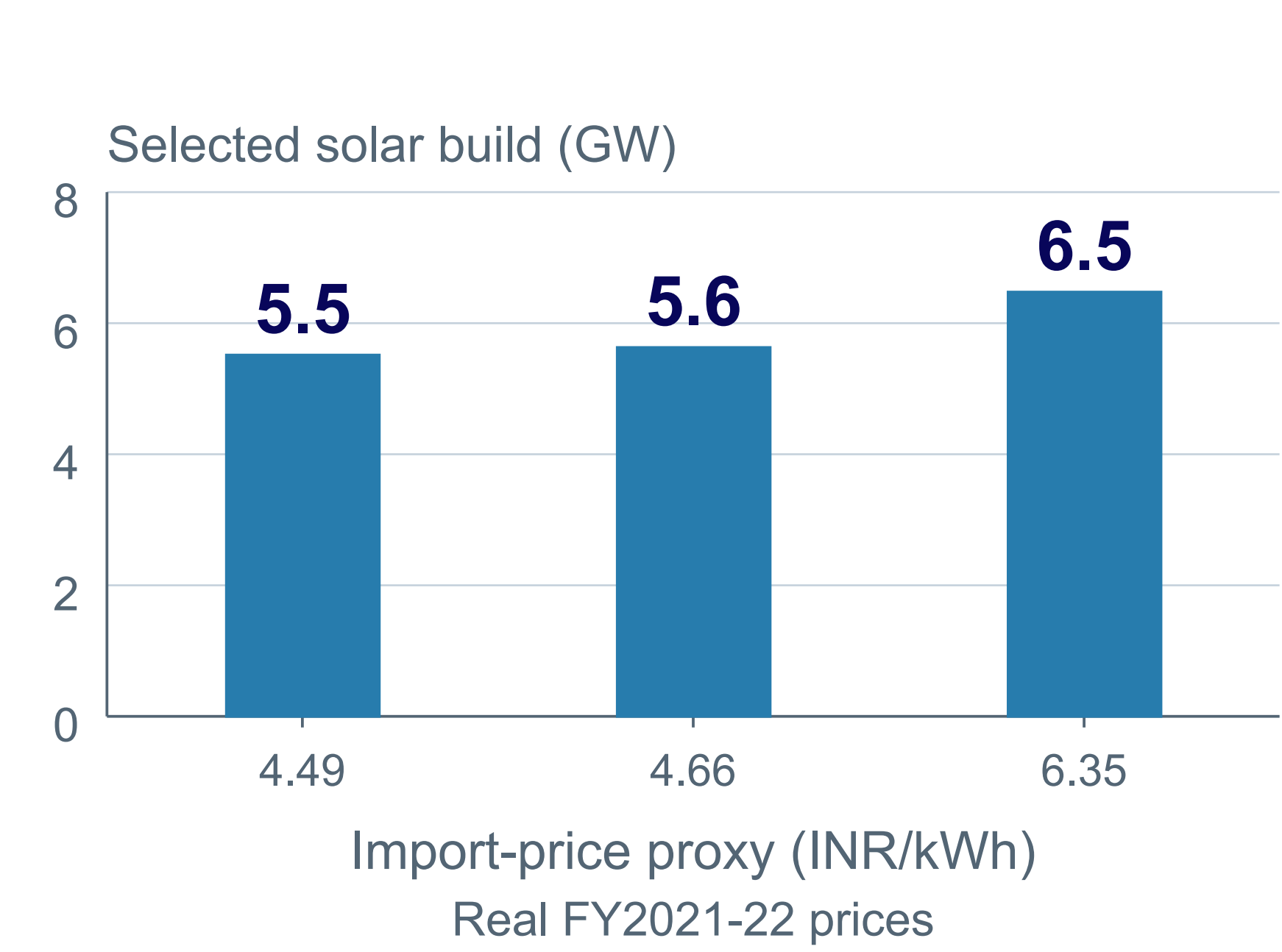
$S_{d+1} = S_d + W_d - E_d/\kappa - R_d$
S storage; W net water; E hydro energy; R extra release; $\kappa = 1,470 \text{ MWh/Mm}^3$.

W is reconstructed, not measured inflow.

Analysis and assessment

D. Import-price sensitivity

Partial economics v1.0 [4]

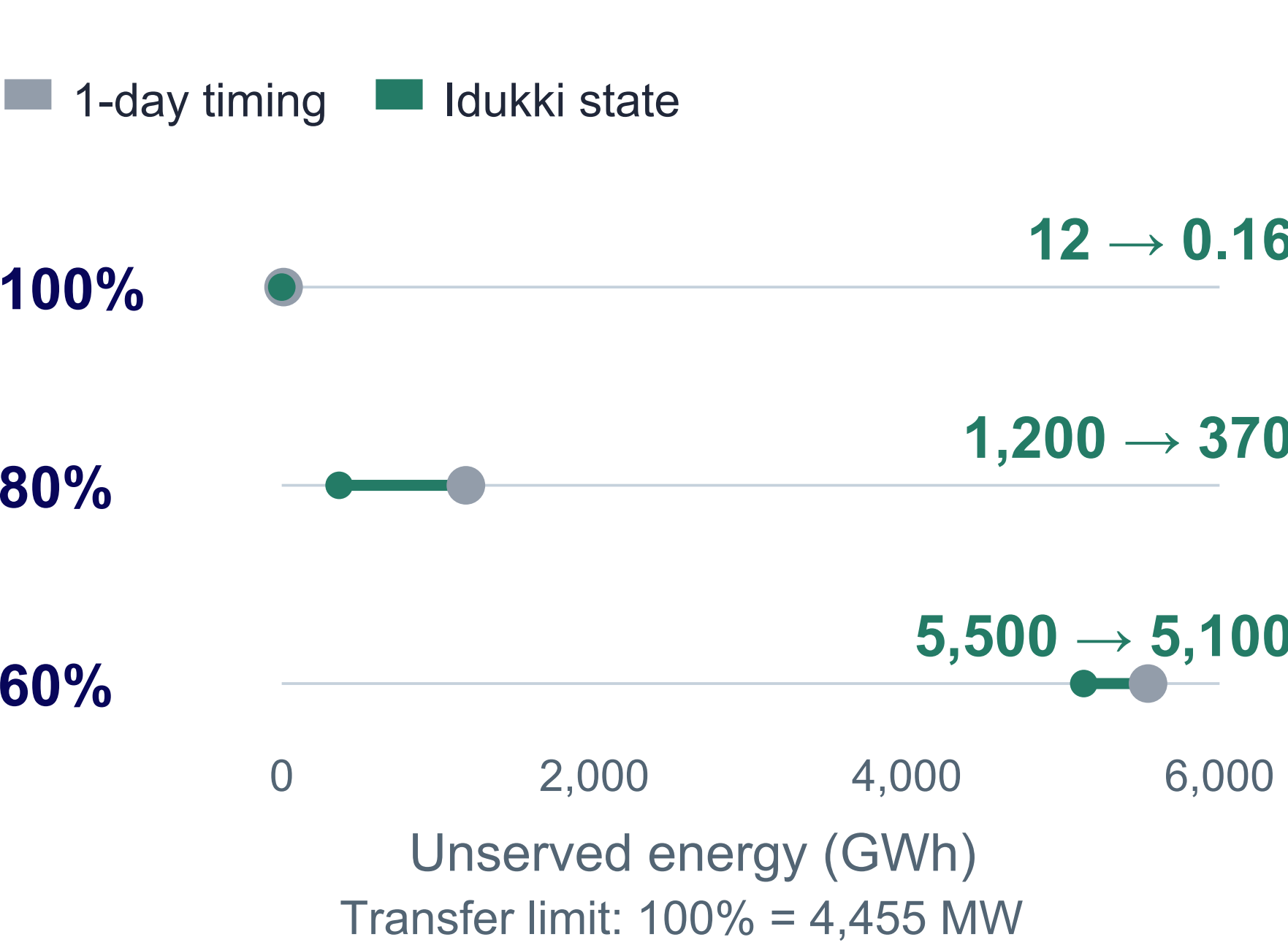


Higher import prices select more solar. Lower demand, full transfer, high renewables and low battery cost.

Partial costs; not project finance.

E. Stateful reservoir model

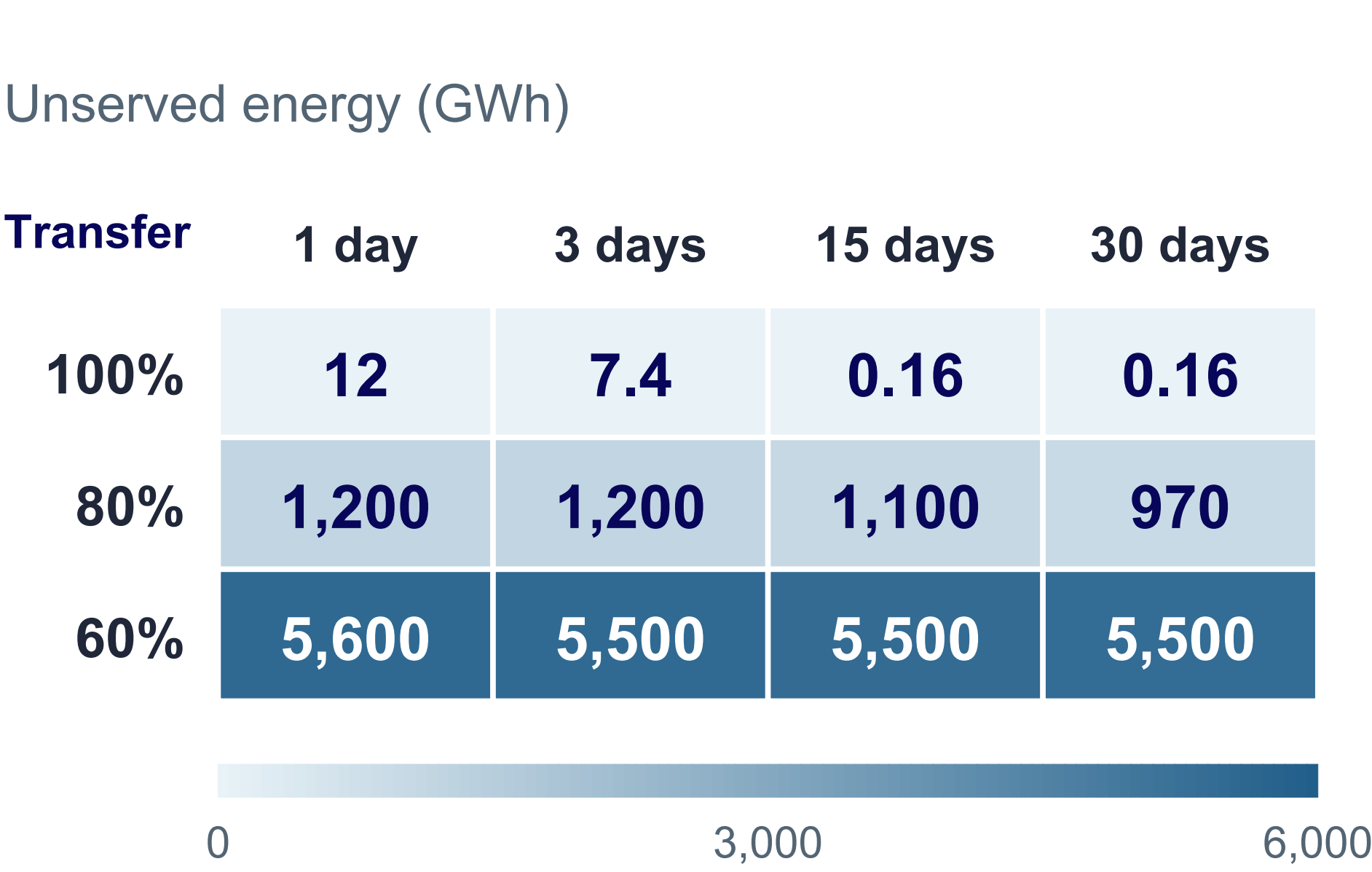
364 days / 8,736 hours · v1.3 [5]



69% less shortage at 80% transfer. FY2030-31 reference demand and renewables; full Idukki; low battery cost; KSEBL price.

F. Hydro timing windows

365 days / 8,760 hours · v1.2 [5]



Longer windows give limited relief under deep transfer stress. Reference demand; full hydro; annual energy fixed at 7.43 TWh.

Windows are not reservoir durations.

Conclusions

Reservoir flexibility reduces shortage, but **5.1 TWh remains unserved at 60% transfer** in the stateful pilot (E).

Assess timing and interconnection together.

Limitations and next steps

Validation: inflow, releases and efficiency remain unvalidated; 11 generation and 11 storage gaps interpolated in v1.3 only.

Next: admit official KSEB records and catchment evidence before physical validation [6].

References and reproducibility

[1] Kerala SLDC daily statistics, FY2024-25.
[2] GSA2, 1999-2018. [3] WP6 synthetic EV dispatch.
[4] PyPSA economics v1.0. [5] Hydro v1.2 / Idukki v1.3.

[6] KSEB v1.5: workbook acquisition blocked. LRIS catchment PR105: under review; catchment not admitted (27 Sep 2026).
Full-precision data, assumptions and code: [kerala2040.github.io/#data](https://github.com/AbhijithSivaprasadan/kerala2040)